Ashween Ramakrishnan

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LinkedIn | GitHub | Website | Google Scholar

SUMMARY

Software Engineer with nearly four years of experience delivering scalable full-stack and AI-driven solutions across health, education, and enterprise platforms. Passionate about building products that improve digital experiences and drive real-world impact. Skilled in applying machine learning, web development, and intelligent automation to solve complex problems. Known for strong problem-solving skills, adaptability, and effective cross-functional collaboration.

EDUCATION

The University of Adelaide (now part of Adelaide University) — **Adelaide, Australia**
Master of Computer Science | GPA: 6.73/7.0 — Jul 2024 - Jul 2026

Amrita Vishwa Vidyapeetham — **Coimbatore, India**
Bachelor of Technology in Computer Science and Engineering — Jul 2016 - Jun 2020

SKILLS

- **Programming Languages:** Python, Java, JavaScript, PHP
- **AI / ML & Data:** PyTorch, OpenCV, Scikit-Learn, LangChain, Hugging Face, ChromaDB
- **Web & Frameworks:** React, Vue.js, Django, Node.js, Spring, Hibernate, Laravel, Nuxt.js, HTML, CSS, jQuery
- **Databases & Cloud:** MySQL, MongoDB, AWS S3, REST APIs
- **DevOps & Developer Tools:** Docker, Jenkins, Git, SVN, Jira, Kubernetes, Ansible

EXPERIENCE

Allied AU - Software Engineer Intern — Jul 2025 - Oct 2025

- Enhanced core features of the Allied AU healthcare platform, improving **email notification workflows (AWS SES)**, **Practitioner Calendar scheduling**, **Referrer-Client Bookings**, and **advanced search modules** to deliver a smoother experience for practitioners and clients.
- Resolved critical issues in **sign-up**, **booking**, and **NDIS workflows**, fixing navigation errors, placeholder logic, and validation conditions to improve onboarding efficiency and platform accessibility.
- Contributed to **Sole Trader** and **business onboarding features**, implementing practitioner availability management, calendar integration, and UI/UX consistency improvements to streamline practitioner operations.
- Diagnosed and resolved **frontend (Nuxt.js)** and **backend (Laravel) issues**, collaborating with QA and DevOps teams to support CI/CD pipelines, staging deployments, and reliable release cycles.
- Participated in **bug triage**, **staging validation**, and **release readiness reviews**, helping prioritise high-impact issues and support timely delivery of platform features aligned with business and regulatory requirements.

LinkedIn - Software Engineer/Contingent Worker — Jul 2022 - Jun 2024

- Developed a **QR code watermarking system** using JavaScript and jQuery that embeds user-specific identifiers on test pages to deter cheating, reducing reported policy violations by **50%**.
- Built and enhanced **custom checkout** and **invoice workflows** for the Trailhead Academy LMS using **Java**, **JavaScript APIs**, and **CyberSource**, including coupon handling, validation logic, and automated purchase/refund processing to improve transaction reliability and revenue tracking.
- Implemented **advanced course filtering functionality** supporting multi-value and comma-separated filters, along with empty-value handling, significantly improving search flexibility and course discoverability for users.
- Designed backend utilities, REST APIs, and microservice-style modules to **manage custom objects**, **user data deletion**, and **bulk data exports**, enabling secure customer offboarding and improving internal operational efficiency through automated export scripts and optimised system workflows.

- Improved system performance and reliability by **debugging runtime query logs**, **optimising backend methods**, and **implementing subscription validation logic** (event capacity checks, waitlist support), ensuring accurate enrolment and event management.

EduBrite Systems, Inc. (now part of LinkedIn) - Full Stack Developer Oct 2020 - Jun 2022

- Developed **custom LMS web modules** for enterprise clients, including Salesforce and VMware, using Java, Spring, Hibernate, Velocity, JavaScript, HTML, and CSS, improving platform functionality and user experience.
- Designed and implemented a **Java-based email service** that supports personalised emails, reusable templates, default template helpers, and enum-based configuration to automate notifications across workflows.
- Optimised **MySQL database queries** and **schema structures**, reducing query execution time by **30%** and improving system responsiveness for high-volume platform usage.
- Implemented new **learning management features**, including lesson bookmarking, attendance tracking for Salesforce training programs, and event grouping by location to improve course organisation and navigation.
- Improved platform reliability by developing **backend helper methods**, **filtering**, **DocuSign invoice integration**, **bug fixes**, and **codebase cleanup**, while supporting testing and feature enhancements across multiple modules.

ICU Medical India LLP - Python Developer Intern Feb 2020 - Jun 2020

- Developed **Python-based automated test suites** using **Pytest** to validate critical functionalities of the Plum 360 Smart Infusion Pump, improving the reliability of safety-critical medical software used in clinical environments.
- Automated **70+ functional test cases** covering infusion pump features like bolus delivery, multi-step infusion, delayed start, piggyback therapy, flush operations, and alarm handling to ensure accurate medication delivery.
- Designed comprehensive **test scenarios for both nominal** and **edge cases**, validating **communication APIs**, **UI widget interactions,** and therapy programming workflows within a **Linux-based testing environment**.
- Increased **automated test coverage** from **58%** to **90%** by implementing scalable Pytest modules, improving defect detection, and strengthening product safety and regulatory compliance for medical devices.

PROJECTS

- **Smart Contract Vulnerability Detector:** Developed a **fine-tuned LLM classifier** (91.6% accuracy) detecting reentrancy vulnerabilities in smart contracts. Built an AST **control-flow** visualiser and integrated the end-to-end real-time security threat analysis tool into **Adelaide University's CyberLab** platform.
- **Human-Aligned LLM Optimisation (RLHF):** Built an enterprise-grade, end-to-end **RLHF** (Reinforcement Learning from Human Feedback) pipeline using Small Language Models (SmolLM2/Qwen) with **LoRA** fine-tuning, training **reward models**, and optimising **PPO** (Proximal Policy Optimisation) **policy iterations** to ensure high-quality outputs and align AI responses with human safety and preference guidelines.
- **Cybersecurity Intelligence System (RAG):** Developed a Retrieval-Augmented Generation (**RAG**) pipeline using a quantised **Mistral-7B LLM** and **ChromaDB** vector store, scraping **MITRE ATT&CK** enterprise data to generate grounded threat mitigations while reducing hallucinations and evaluating outputs via **RAGAS metrics**.
- **LiveWell – Elderly Care Companion**: Built a mobile companion app featuring geofencing safety alerts, accelerometer-based fall detection, and a **LangGraph, Flask, OpenAI Whisper,** and **Llama-powered** chatbot assistant built with **React Native** for **voice-driven,** personalised support and caregiver notifications.
- **Regex Pattern Matching Using LLMs**: Built a **Django + React** web application that converts **natural-language queries** into regex patterns using **LLMs**, enabling automated text detection and replacement in large-scale CSV/Excel datasets for end-to-end data cleaning and preprocessing pipelines.

CERTIFICATIONS, CONTRIBUTIONS & LEADERSHIP

- **Certification:** **Microsoft Certified: Azure Fundamentals**, **Docker Foundations Professional Certificate**
- **Open-Source**: **OpenSeadragon** - Contributed to the OpenSeadragon open-source image viewer by enhancing **zoom control APIs** and **DOM event handling**, with multiple PRs merged into the core project.
- **Master's in Technology Student Association** (**MiTSA**): Director of Events & Officer of Sports and Wellbeing

— References available on request —